



Genotoxin Stress and the Role of Alkaloids 15

Vaishali Yadav and Divya Vohora

Abstract

Humans are subjected to an array of external and internal stimuli during their lives, which injure genetic material and generate somatic mutations that lead to accumulating genomic stress. Several human pathologies like chronic degenerative diseases, including hepatic, neurodegenerative, and cardiovascular disorders; diabetes; arthritis; cancer; chronic inflammation; and aging, are associated with genotoxic stress. In the past few decades, a great deal of research has been done to find substances that can shield humans from DNA damage and its impact. Natural substances continue to show a beneficial impact on genome integrity. Scientists have identified and reviewed unique bioactive plant-derived compounds, such as flavonoids, phenolic compounds, and carotenoids, that can mitigate the impact of genotoxic stress; however, the genoprotective potential of alkaloids is not being reviewed so far. Herbs high in alkaloidal content have a plethora of pharmacological traits, which are utilized as drugs, and are significant to the pharmaceutical sector. This chapter summarizes the literature on the genoprotective/antigenotoxic action of alkaloids acting against endogenously and exogenously produced genetic damage. The current knowledge regarding the function of alkaloids to guard the genome can be exploited to create dietary plans and natural health products that are beneficial in the sequel of chemotherapeutic-associated toxicities.

V. Yadav

Neurobehavioral Pharmacology Laboratory, Department of Pharmacology, School of Pharmaceutical Education and Research (SPER), Jamia Hamdard, New Delhi, India

D. Vohora (✉)

Neurobehavioral Pharmacology Laboratory, Department of Pharmacology, School of Pharmaceutical Education and Research (SPER), Jamia Hamdard, New Delhi, India

e-mail: dvohra@jamiahamdard.ac.in

© The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2023

R. Tulsawani, D. Vohora (eds.), *Adaptation under Stressful Environments through Biological Adjustments and Interventions*,

https://doi.org/10.1007/978-981-99-7652-2_15

215

Keywords

Genotoxicity · Genoprotective · Anti-genotoxicity · Alkaloids · Piperine

15.1 Introduction

Deoxyribonucleic acid (DNA) in living organisms is constantly exposed to harmful substances. It is assessed that every cellular component in the human body experiences millions of genomic lesions per day (Luca et al. 2016). Genotoxic stress can be defined as a situation that initiates DNA damage compromising the cellular genomic integrity. The capacity of various substances to cause harm to genetic material is known as genotoxicity, while genome instability refers to a series of events associated with an elevated rate of DNA damage and associated mutations, arising from defects in the DNA damage response, DNA double-strand breaks, DNA replication, cell cycle control, or chromosome segregation (Coates et al. 2005; Kovalchuk 2021). Genotoxic stress can be induced by intrinsic processes (such as reactive oxygen species generated during physiological and pathogenic progressions) and extrinsic compounds (such as UV light, ionizing radiation, anti-cancer drugs, and even air, water, and food pollutants). Human exposure to numerous genotoxic stressors has escalated considerably in recent years. Specifically, cancer patients undertaking radiotherapy and chemotherapy are overexposed to cyto(genoto)xic agents. Furthermore, several pathological conditions exhibit genome instability events and DNA damage, such as aging, carcinogenesis, immune deficiencies, infertility, cardiovascular diseases, metabolic syndrome, and neuromuscular and neurodegenerative diseases like Alzheimer's disease (Georgoulis et al. 2017). Therefore, it is of significant interest to identify compounds, particularly from traditional medicinal plants, that can guard against genotoxic harm and its consequences.

Several reviews related to the anti-genotoxic activity of phytochemicals like flavonoids, polysaccharides, phenolic compounds, and carotenoids are already available in the public domain (Hosseinimehr 2010; George et al. 2017; Izquierdo-Vega et al. 2017; López-Romero et al. 2018); however, we did not find any such information regarding genoprotective action of alkaloids. In view of the above, this chapter summarizes research studies on the genoprotective/antigenotoxic effects of alkaloids against both endogenous and exogenously induced genetic damage, specifically, DNA damage induced by various carcinogenic drugs. Further, this chapter also sheds light on sources of genotoxic stress, physiological response to genotoxic stress, general methods to detect DNA damage, the purpose of pharmacological interventions, and the role of specific alkaloids in the same.

15.2 Source and Types of Genotoxic Stress

A cell experiences thousands to lakhs of molecular lesions per day as a result of external influences and regular metabolic activities. Although this only accounts for 0.000165% of the roughly 6 billion (3 billion base pair) bases that make up the human genome, if left unrepaired, it could lead to mutations in important genes (such as tumor suppressor genes), which would significantly increase the risk of tumor development and disease states like cancer (Luca et al. 2016).

There are various types of DNA damage that can happen as a result of either regular cellular functions or cells being exposed to external factors that cause DNA damage. DNA bases can be damaged by: (1) Oxidative processes, (2) Alkylation of bases, (3) Base loss caused by the hydrolysis of bases, (4) Bulky adduct formation, (5) DNA crosslinking, and (6) DNA strand breaks, including single- and double-stranded breaks. Table 15.1 represents DNA damage inducers from various sources. Below is a description of genetic lesions induced by several factors:

15.2.1 Oxidative Damage

Reactive oxygen species (ROS) have the ability to produce substantial physiological stress and harm cells, comprising DNA oxidative injury. Among the most susceptible and electrophilic reactive oxygen species are the hydroxyl radicals ($\cdot\text{OH}$), which can be created by UV and ionizing radiations as well as other radicals generated during enzymatic activities. Hydroxyl radicals subsequently react with nucleobases, as guanine has the lowest ionization potential, it is quickly oxidized to form 8-oxo-7,8-dihydroguanine (8-oxoG); a major prevalent genomic lesion and a potential hallmark of oxidative stress (Cooke et al. 2003).

15.2.2 Alkylation of Bases

Alkylating substances are abundant in ecosystems and are generated internally by cells as waste products of metabolism. They cause cytotoxic lesions, and mutagenic or neutral lesions to be incorporated into genomic bases. Cytotoxic lesions inhibit the genomic replication process, interfere with the transcription process, or activate the cascade for apoptosis, while mutagenic lesions lead to miscoding and mutations in a newly synthesized genome. The most common type of alkylation is methylation with the major products including N7-methylguanine (7meG), N3-methyladenine (3meA), and O6-methylguanine (O6meG) (Kondo et al. 2010).

15.2.3 Base Loss

When the Nitrogen-glycosyl bond connecting the nitrogenous base and the sugar-phosphate backbone is disrupted by water molecules spontaneously or is broken by a

Table 15.1 Compounds inducing genotoxic stress and their DNA-damaging effects

S. no.	Compounds	DNA damaging action	Type of DNA damage	References
Environmental toxicants				
1	Exposure of Ultraviolet B radiation	Formation of dimeric DNA photoproducts, specifically cyclobutane pyrimidine dimers and pyrimidine-6,4-pyrimidone photoproducts	DSSB and DDSB	Chatterjee and Walker (2017)
2	Gamma rays	ROS formation	DSSB and DDSB	
3	Hydrogen peroxide	Formation of hydroxyl radicals via a Fenton-type reaction	DSSB, DDSB, and CA	
4	Polycyclic aromatic hydrocarbons	Activation of the aryl hydrocarbon receptor	DNA adduct formation	
5	Styrene	Generation of electrophilic compound: styrene-7,8-oxide	DSSB and DDSB	
6	N-nitrosamines	Induction of CYP450	DNA adduct formation	
7	Heterocyclic aromatic amines	Inducer of CYP450 1A superfamily, N-acetyltransferases, and sulfotransferases	DSSB, DDSB, CA, and SCE	
8	Heterocyclic aromatic nitro derivatives	Inducer of cytochrome-P450	DSSB, alkali-labile sites formation, and CA	
9	Asbestos	Generation of ROS	DSSB, DDSB	
Chemotherapeutic agents				
10	Alkylating agents (nitrogen mustards)	Base alkylation-monofunctional DNA adducts	Inhibits DNA replication process	Swift and Golsteyn (2014)
11	Platinum drugs (cisplatin, carboplatin, and oxaliplatin)	Monofunctional DNA adducts	Inhibits DNA replication process	
12	Topoisomerase poisons (Camptothecin and etoposide)	Blocks topoisomerase enzymes in DNA	DSSB and DDSB	
13	Antimetabolites (5-fluorouracil)	Misincorporates into DNA	Inhibits DNA replication process	

DSSB DNA single-strand breaks, *DDSB* DNA double-strand breaks, *ROS* reactive oxygen species, *CA* chromosomal aberrations, *SCE* sister chromatid exchanges

genomic glycosylase to produce an intermediate in the base excision repair pathway, abasic or AP (apurinic/apyrimidic) sites are continually generated in the DNA. These are fundamentally unstable and easily undergo a beta-elimination reaction that destroys the phosphodiester link of the remaining deoxyribose, resulting in single-strand breaks (Chatterjee and Walker 2017).

15.2.4 Bulky Adduct Formation

Certain substances are physiologically responsive and generate covalent bonds with biological macromolecules like DNA and proteins, which will result in the development of big, bulky appendages, or adducts, that split off from the original molecule. For example, benzo[a]pyrene is exposed to humans via cigarette smoking, activated into (+) benzo[a]pyrene-7,8-dihydrodiol-9,10-epoxide, which generates a DNA adduct with guanine residues and disrupts the genomic backbone (Poirier 2012).

15.2.5 DNA Crosslinking

When cross-bridging chemicals covalently join two nucleotide residues from the same genomic strand (intra-strand) or from the opposing strands (inter-strand), they cause DNA crosslinking damage. The nucleotide excision repair mechanism is a simple and effective method, undergoing in a physiological state, for removing intra-strand crosslinks. However, DNA strand detachment is completely prevented by inter-strand crossing, which obstructs crucial DNA metabolic functions including transcription and replication. Further, this can be exceedingly harmful if not corrected, especially in dividing cells, as it can halt DNA replication and result in cell death. Chemotherapeutic agents like melphalan and cisplatin are examples of inter-strand cross-forming compounds (Lehmann and Wennerberg 2021).

15.2.6 DNA Strand Breaks

DNA double-strand breaks (DSBs) can occur as a consequence of endogenous mechanisms like DNA replication and repair along with external factors like radiation and certain chemicals. DSBs are especially problematic as, when left unrepaired, they can result in cell death. Additionally, incorrectly repaired DSBs can result in DNA deletions, translocations, and fusions. These outcomes are referred to as genomic rearrangements collectively, and they are frequently observed in malignant cells (Chatterjee and Walker 2017).

15.3 Physiological Response to Genotoxic Stress: DNA Damage Response

The process by which cells react to genotoxic stress is extremely complicated. However, it is explained as a series of activities, sequentially engaged to cause the DNA damage response (DDR) after recognition of genotoxic stress (Mohanani et al. 2021). It involves detecting genomic damage via a variety of sensor molecules, engaging on checkpoint proteins, and then activating repair mechanisms. The elements of DDR encompass “sensors,” “transducers,” and “effectors”; these groups mediate final outcomes such as repair and/or apoptosis. Genomic injury/lesions serve as an initial signal, which is perceived by “sensors,” and escalated via “transducers.” The “effectors” ultimately catch the signal and carry out various cellular processes (Yang et al. 2004).

ATM and ATR-Dependent DNA Damage Response The members of the phosphatidylinositol 3-kinase (PI-3) superfamily are stimulated at the initial phase of the DDR, which might function as sensors, as well as initiators of the physiological genotoxic stress response, including ataxia-telangiectasia mutated (ATM) and ATM and RAD3 (ATR) activation. ATM and ATR can be activated by interaction with genomic lesions or by interaction with DNA repair proteins (Yang et al. 2004). The involvement of ATM and ATR is well studied in DNA double-strand break formation. Mre11–Rad50–Nbs1 complex recognizes this injury, following the recruitment of ATM. The active ATM subsequently initiates two processes, which are crucial for checkpoint signaling and DSB repair, that organize the chromatin and handle the DNA close to the DSB lesion. The very first processing of ATM is phosphorylation of histone H2AX, followed by activation of Mdc1 and formation of protein complexes encompassing Mre11–Rad50–Nbs1 complex, p53-binding protein, BRCA1, and ATM. This is followed by the targeting of ATM toward substrate-like kinase checkpoint kinase-2, which results in the dispersal of injury signaling across the nucleus (Shiloh 2006; Li and Zou 2005). The chromatin alteration signaling is widespread in the cell cycle, independent of cyclin-dependent kinase activity, while ATR stimulation is cyclin-dependent kinase-dependent, which is constrained to S and G2 phases. Double-strand breaks expand the single strand of DNA, which is covered by replication protein A complex, an intermediate for DNA repair and ATR-dependent signaling. Further, ATR is activated following independent recruitment of the ATR-ATR-interacting protein complex and DNA topoisomerase 2-binding protein 1. The effector signaling kinase, checkpoint kinase-1, whose activation also needs Claspin and ATR-dependent phosphorylation of Rad17, is one of the substrates that ATR then targets. The ATM-Chk2 and ATR-Chk1 transmission molecules then cooperatively orchestrate the checkpoint responses and DNA repair in the damaged cell (Bartek and Lukas 2007; Zhou and Elledge 2000; Sirbu and Cortez 2013). A schematic presentation of the genotoxic stress response is presented in Fig. 15.1, as a simplified version of the “p53 signaling pathway” obtained and developed from Kanehisa Laboratories in the Institute for Chemical Research, Kyoto University. They are dedicated to establishing KEGG (Kyoto

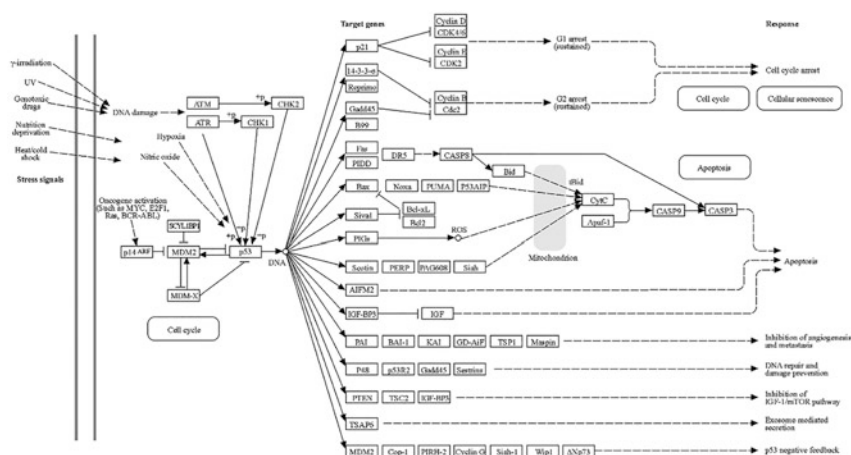


Fig. 15.1 Physiological response to genotoxic stress: The figure represents sources of genomic injury (such as radiations and genotoxic drugs) which generates a range of genomic lesions (like oxidative DNA damage, DNA adduct, alkylation of the genome, and DNA single- and double-strand break formation), these lesions when left unrepaired cause a severe malfunction in DNA machinery consequently leading to cell death. These genomic lesions are sensed by the members of the phosphatidylinositol 3-kinase (PI-3) superfamily, mainly including ataxia-telangiectasia mutated (ATM) and ATM and RAD3 (ATR) activation. Stimulation of ATM and ATR further leads to the recruitment of Chk1, Chk2, and p53. These proteins via the involvement of various genes carry out the final cellular responses like control of the cell cycle, apoptosis, DNA repair, and damage prevention. The figure is developed by the Kanehisa Laboratory (now part of the Bioinformatics Center) in the Institute for Chemical Research, Kyoto University

Encyclopedia of Genes and Genomes), a tool to derive cellular response from its genomic data (Kanehisa et al. 2004).

15.4 Methods to Detect DNA Damage

The Organization for European Cooperation and Development (OECD), US Environmental Protection Agency (EPA), and International Council for Harmonization (ICH) have given various guidelines to predict genotoxicity, which is an essential step to be followed during the drug development process of any substance. In the pharmaceutical industry, genotoxicity testing, toxicology, and the detection of genomic abnormalities in clinical specimens depend on the precision/accuracy of the method to recognize DNA damage.

In context to this chapter, the preventive or damaging effects of alkaloids can be evaluated in the following system:

1. Cell-free medium like plasmid relaxation assay.
2. Prokaryotes-based test system, which consists of SOS chromotest, Umu assay, and Vitotox assay.

3. Yeast-based test system.
4. Higher plant-based genetic method consisting of *Allium cepa* assay.
5. Cytogenetic testing strategies such as micronucleus assay (cytokinesis block micronucleus assay-CBMN and in vivo rodent micronucleus assay), chromosome aberration test (in vitro and rodent bone marrow chromosome aberration test), and sister chromatid exchange assay.
6. Unicellular-based methods including comet assay.
7. DNA repair-centered strategies such as unscheduled DNA synthesis (UDS) assay.
8. DNA fragmentation tests like TUNEL assay, DNA ladder assay, annexin V Staining, and diphenylamine (DPA) assay.
9. Biomarker-based strategies, which evaluate 8-Hydroxy-2-deoxy guanosine (8-OHdG), γ H2AX, and DNA adduct detections.
10. Cell cycle alterations, detected via flow cytometry.
11. Genetic/molecular strategies like DNA microarray, polymerase chain reaction (PCR), and random amplification of polymorphic DNA (RAPD), and
12. High throughput/advanced strategies like 3D cell culture test system and fluorescence in situ hybridization (FISH) test system (Figuerola-González and Pérez-Plasencia 2017; Mohamed et al. 2017; Turkez et al. 2017; Nelson et al. 2017; Reddig et al. 2018; Firsanov et al. 2021).

15.5 Purpose of Pharmacological Intervention

An agent that combats genotoxic stress has the ability to stop, block, or avoid the spread of genetic damage. The development of efficient anticancer drugs is drawing increasing attention as cancer is the main cause of human mortality, accounting for around 9.6 million deaths worldwide in 2018. While chemotherapy and radiotherapy are the pillars of cancer treatment, their significant adverse reactions frequently limit their effectiveness and utilization. Most chemotherapy drugs are genotoxic, which are likely to cause epigenetic and genetic damage. Chemotherapeutic agents like cyclophosphamide, cisplatin, 5-fluorouracil, bleomycin, doxorubicin, methotrexate, etc., are the cornerstones in cancer treatment, which are designed to directly target DNA and are associated with high levels of DNA damage. With the significant rise in the number of cancer patients each year, it is predicted that by 2040, there will be 15.3 million cancer patients to require first-line chemotherapy treatment (Sak 2012; George et al. 2017). Thus, it is essential to search for compounds that can counter the genotoxic damage of such drugs. The floral world is composed of plants with a wide range of chemical properties, most of which have a favorable toxicity profile. The most commonly utilized sources of phytochemicals for cancer therapy are herbal formulations, plant crude extracts, and even food. These have demonstrated unmatched benefits in minimizing the side effects of anticancer medicines (Aprotosoiaie et al. 2019; Liu et al. 2021). These phytochemicals may function as adaptogens, shielding the body from the harmful consequences of intervention, or as biologic response modifiers, adding to or even enhancing the effects of

pharmacological medicines. Further, numerous genome-protecting substances are emerging as putative genoprotectors and have been linked to the management of other human pathologies like neurodegeneration and neuromuscular disease. As a result, researchers have generated a renewed interest in this area and broadened the horizon of such compounds (Proshkina et al. 2020).

Alkaloids and other phytochemicals might show protection against genotoxic stress by following mechanisms: (a) suppression of genotoxin uptake; (b) complexation or deactivation of toxin; (c) blocking and competition mechanisms such as scavenging of reactive oxygen species and protection of DNA nucleophilic sites; (d) alteration in transmembrane transport; (e) alteration in DNA repair mechanism; and (f) alterations in xenobiotic metabolizing enzymes (López-Romero et al. 2018; Morales-González et al. 2019; de Kok et al. 2010).

15.6 Alkaloids Against Genotoxic Stress

Several bioactive chemicals found in nature can be transformed into the medications we use today. Most chemicals, which help plants thrive and propagate, are recognized to be synthesized by plants. Alkaloids, phenols, steroids, glycosides, tannins, terpenoids, and phytoalexins are examples of such bioactive metabolites. These metabolites display biodynamic behaviors that are beneficial to both human and animal well-being. Alkaloids are amid the immense class of natural compounds, which constitutes a plethora of chemical structures, including over 12,000 entities. The main prerequisite for being classified as an alkaloid is the presence of a basic nitrogen atom in any location throughout the molecule, which excludes nitrogen found in peptides (Debnath et al. 2018). Alkaloids are universally scattered in the plant kingdom and belong to families like Ranunculaceae, Apocynaceae, Leguminosae, Erythroxylaceae, Papaveraceae, Amaryllidaceae, Menispermaceae, Loganiaceae, etc. Alkaloids are typically categorized into (a) true alkaloid, nitrogen-containing heterocyclic ring, for instance, cocaine, morphine, quinine, etc. (b) Protoalkaloids, where nitrogen is not a part of heterocyclic ring, for instance, yohimbine, mescaline, and hordenine; and (c) pseudoalkaloids, where they are not derived from amino acids, for instance capsaicin, caffeine, and ephedrine. Further, alkaloids can also be categorized depending on their chemical structure such as tropane alkaloids (example: cocaine), pyrrolizidine alkaloids (example: senecionine), piperidine alkaloids (example: lobeline), quinolines alkaloid (example: quinine), isoquinoline alkaloid (example: berbamine), indole alkaloids (example: ergotamine), steroidal alkaloids (example: jervine), purine alkaloids (example: caffeine), and pyrrolidine alkaloids (example: hygrine) (Dey et al. 2020). Herbs high in alkaloidal content have a plethora range of pharmacological traits. Alkaloids have been used as drugs throughout human history, which highlights their significance to the pharmaceutical sector. The below mentioned are the alkaloids that have shown evidence to be utilized against various genotoxic stress. The chemical structure of these alkaloids is presented in Fig. 15.2, while Table 15.2 represents the specific details of the experiments conducted on them.

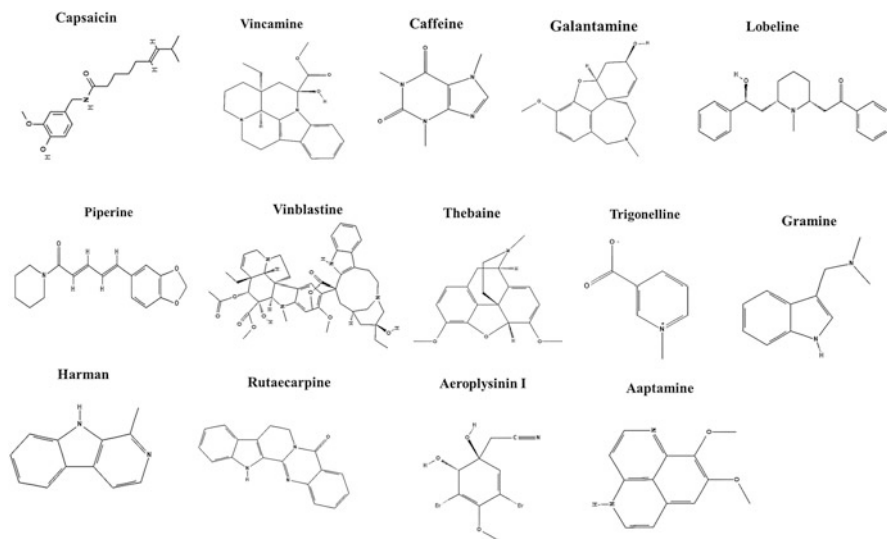


Fig. 15.2 Chemical structure of various alkaloids acting against genotoxic stress

15.6.1 Capsaicin

The main spicy alkaloid present in chili peppers is referred by capsaicin, a homovanillic acid derivate, which is chemically 8-methyl-*N*-vanillyl-*trans*-6-nonenamide. The chili pepper is the fruit of the herbaceous, Solanacea family, endemic to the Americas, Capsicum genus plants. Since Mesoamerican civilizations cultivated chilies, they have been used in food and medicine all throughout the world. A multitude of human disorders, including chronic pain, obesity, diabetes, cardiovascular conditions, respiratory diseases, and cancer have been identified as possible therapeutic targets for capsaicin. Capsaicin is commercialized as topical analgesic cream, used for the treatment of medically diagnosed painful diabetic neuropathy or postherpetic neuralgia. Conflicting results have been found on the genotoxicity/genoprotective potential of capsaicin. Díaz Barriga Arceo et al. (1995) have demonstrated that a low dose of capsaicin (1.46 mg/kg) resulted in genomic injury after 32 days of consecutive intraperitoneal administration to mice, while at a higher dose (1.94 mg/kg), the injury occurred from day 16 onward, as observed by calculating the frequency of micronucleated normochromic erythrocytes. Lewinska and coworkers (2015) have also demonstrated the induction of DNA double-strand breaks in human prostate carcinoma cells and human lung carcinoma cells by capsaicin at a concentration of 100 μ M. Contrary to these, it was reported that capsaicin did not show any genomic lesions up to the concentration of 250 μ M in human endothelial cells, as evaluated by comet assay, while stimulation of inflammatory cascade was observed (Richeux et al. 2000). Caetano et al. (2018) further explored the role of capsaicin in the 1,2-dimethyl-hydrazine-associated colon carcinogenesis model of rats, where it resulted in the protection of genomic injury.

Table 15.2 Protective effect of alkaloids against various genotoxins

S. no	Alkaloid	Inducer of genotoxic stress	Analysis system	Evaluating parameters	Result	References
1	Aaptamine	Cisplatin	Rat kidney proximal tubular cells and human bladder cancer cells	γ H2AX, p53, and checkpoint kinase-1	Geno-protection observed in tubular and glomerular rat kidney cells	Funk et al. (2014)
2	Aaptamine and Aeropysinin-1	Daunorubicin	Acute myeloid leukemia	γ H2AX, Kap-1, and Chk-1	Reduction in DNA damage response	Stuhldreier et al. (2015)
3	β -carboline alkaloids (Harmine, Harmol, Harmaline, and Harmatol)	Hydrogen peroxide; 100 μ M	Chinese hamster lung fibroblasts (V79) cells	Comet assay	Hydroxyl radical-scavenging action is responsible for genome protection	Moura et al. (2007)
4	Caffeine	Vinblastine (1.5 mg/kg)	Swiss male albino mice	Mitotic metaphase chromosome aberration assay and micronucleus test from bone marrow	A reduction in cytogenetic markers was observed	Geriyol et al. (2015)
5	Capsaicin	1,2-Dimethylhydrazine	Male Wistar rats	Leukocyte and fecal water genotoxicity	Reduced cyto- and genotoxicity, promotes antiproliferative and proapoptotic responses	Caetano et al. (2018)
		Mitomycin C	Albino rats	Micronucleus test and chromosomal aberrations	Absences of genomic lesions	Raymond et al. (2004)
6	Gramine	2-Aminofluorene (20 μ g/plate), sodium azide (1 μ g/plate), and 2-nitrofluorene (1 μ g/plate)	Bacteria strain of <i>Salmonella typhimurium</i> and Swiss albino mice	% mutagenic inhibition, bone marrow micronucleus assay and chromosomal aberrations	Genome protection activity due to antioxidant potential	Pathak et al. (2018)

(continued)

Table 15.2 (continued)

S. no	Alkaloid	Inducer of genotoxic stress	Analysis system	Evaluating parameters	Result	References
7	Galantamine	Amyloid beta A β ₍₁₋₄₂₎	SH-SY5Y cell line	Clonogenic assay, cytokinesis block micronucleus cytome and comet assay	Genome protection was observed in neuronal cells	Castillo et al. (2016)
8	Lobeline	Ethanol (2.5 g/kg), hydrogen peroxide, cyclophosphamide	CF-1 male mice	Comet assay, micronucleus test, <i>Salmonella</i> /microsome mutagenicity assay	Genome protection was observed	da Costa e Silva et al. (2014)
9	Thebaine	Pilocarpine	CF-1 male mice	Comet assay, DPH activity, superoxide dismutase and catalase	Genome protection was observed in hippocampus and cortex	da Costa e Silva et al. (2018)
10	Trigonelline	Mitomycin C	TK6 cells	Cytokinesis-block micronucleus test	Genome protection was observed only after co-treatment of toxicant and drug	Reimann et al. (2021)
11	Piperine	UV-B irradiation	Primary Human Dermal Fibroblasts and BALB/C mice	P53, ATM, ATR, γ H2AX, Chk1, and Chk2	Provides genome-protection through PI3K-AKT-Nrf2 pathway	Tanveer et al. (2023)
		7,12-dimethylbenz[a]anthracene	Male golden Syrian hamsters	Formation of micronucleus polychromatic erythrocytes and chromosomal aberrations	Antioxidant action was responsible for genome protection	Balakrishnan et al. (2008)
		Benzo(a)pyrene	Male Swiss albino mice	8-oxo-20-deoxyguanosine, Comet assay	Prevented DNA damage in lungs and liver	Sehgal et al. (2011)

		Cyclophosphamide	Wistar albino rats	Mammalian erythrocyte micronucleus test, diphenylamine assay, quantification of 8-OHdG and γ H2AX	Prevented DNA damage in bone marrow cells, liver, and hippocampus. Reversed the markers of oxidative DNA damage and DNA double-strand breaks	Yadav et al. (2023a, b)
12	Rutaecarpine	Hydrogen peroxide; 500 μ M	Hepa-1c1c7 cells	Comet assay	Reduced DNA-damaged strands	Lee et al. (2012)
		Benzol[a]pyrene; 5 μ M	Mucoepidermoid pulmonary carcinoma cells and hepatocellular carcinoma (Hep3B) cells	Commercial DNA damage kit	Prevention of genomic damage	Li et al. (2019)
13	Vincamine	Cisplatin, 10 mg/kg, i. p.	Albino rats	Urea, creatinine, kidney injury molecule-1, malondialdehyde, myeloperoxidase, nuclear factor erythroid 2-related factor 2, Hemeoxygenase-1, Toll like receptor 4, interferon gamma-CD44, transforming growth factor beta and DNA fragmentation	Anti-inflammatory and antioxidant activity are attributed to prevention in DNA damage	El-Sayed et al. (2021)

Capsaicin at the dose of 50 mg/kg, for 4 weeks (3 times a week) reduced the 1,2-dimethyl-hydrazine-associated genomic damage, as evaluated in rat leukocytes by comet assay. The findings were analogous to fecal water genotoxicity in CaCO-2 cells at 50 mg/kg. The report further established a reduction in the genetic expression of proliferation and proapoptotic markers by capsaicin.

15.6.2 Caffeine

Chemically, caffeine is 1,3,7-trimethyl xanthine, a purine alkaloid, which is obtained from *Camellia sinensis* and *Coffea arabica* commonly referred to as tea and coffee. Studies suggested that caffeine has resulted in genomic damage, when tested in various in vitro (CHO cells, rat kidney cell line, and cultured mouse preimplantation embryo cells) and in vivo systems (albino mice); however, the ability of caffeine to cause genomic damage is observed when the concentration is raised from ten to hundred times more than consumed in humans (Nehlig and Debry 1994). As a result, caffeine is not believed to be genotoxic to animals and does not appear to be mutagenic to people when consumed. Stoilov and coworkers (1994) studied the temperature-dependent effect of caffeine on radiation-associated genomic damage. At a lower temperature, it exhibited genome protective effects, while at a higher temperature, an increased rate of damage was observed in lymphocytes. Caffeine's capability to either exacerbate or decrease radiation-associated genomic damage appears to be correlated with its capacity to bind to DNA and neutralize radicals. Further, the experiment of Geriyol et al. (2015) demonstrated that caffeine administered to mice at the dose of 25, 50, and 100 mg/kg, counteracted vinblastine-associated cytogenotoxicity, as indicated by a decline in chromosomal aberrations in erythrocytes derived from bone marrow. The effect was postulated to be the free radical scavenging property of caffeine. Kumar et al. (2001), have also suggested similar action of caffeine, taking advantage of its antioxidant property, where it efficiently shielded DNA against ultraviolet radiation (at a dose rate of 1.2 Gy/min), when exposed to plasmid pBR322 DNA. Further, to comprehend the antioxidant mechanisms, the reaction of caffeine with various radicals (such as superoxide, hydrogen peroxide, and deoxyribose peroxyl radical) was studied, where caffeine scavenged the primary as well as secondary radicals generated by radiations.

15.6.3 Galantamine

Galantamine can be isolated from bulbs and flowers of *Galanthusnivalis* (Common snowdrop), *Galanthuscaucasicus* (Caucasian snowdrop), *Galanthusworonowii* (Voronov's snowdrop), and some other members of the family Amaryllidaceae, such as *Narcissus* (daffodil), *Leucojumaestivum* (snowflake), and *Lycoris* including *Lycoris radiata* (red spider lily). It is an acetylcholinesterase inhibitor used for improving cognitive functions in Alzheimer's disease patients at mild, moderate,

and advanced moderate stages (Villarroya et al. 2007). The research team of Castillo has recently demonstrated the efficacy of galantamine against genotoxic stress exacerbated by amyloid beta, in traditional assays of cytogenotoxicity such as clonogenic assay, cytokinesis block micronucleus assay, and comet assay in SH-SY5Y human neuroblastoma cell line. The authors speculated that the positive outcome of galantamine is associated with its antioxidant property. Galantamine could possibly protect against genomic injuries by stimulating DNA repair enzymes and the replication process (Castillo and Aristizabal-Pachon 2017; Castillo et al. 2016, 2018).

15.6.4 Piperine

Piperine is chemically an alkaloid, which can be isolated from the *Piper* genus, family Piperaceae. Piperine gained the attention of health professionals because of its wide range of biological actions. Also, humans are regularly exposed to piperine, majorly via diet, in the form of black pepper. An average intake of 333 mg per person of pepper results in an intake of 16.5–29.7 mg of piperine per person per day. Reen et al. (1997) have provided in vitro validation of piperine to counter genomic damage. The authors used genetically engineered V79 Chinese hamster cell constructs of rat liver CYP4502B1, where piperine reversed the genomic damage against aflatoxin B1. The study established that piperine is able to suppress CYP4502B1 action while combating pro-carcinogens (such as aflatoxins) triggered by cytochrome enzymes.

In 2005, Selvendiran et al. conducted a preliminary experiment to assess the genopreventive action of piperine in mice. Piperine when administered to mice at the dose of 75 mg/kg for 5 days, reversed the frequency of micronucleus in erythrocytes of bone marrow after single-dose administration of benzo(alpha)pyrene (25 mg/kg, intraperitoneal administration) and cyclophosphamide (20 mg/kg, intraperitoneal administration). The action was attributed to the antioxidant effects and alteration of phase II enzymes (Selvendiran et al. 2005). Sehgal et al. (2011) assessed the effect of piperine against benzo(a)pyrene-induced injury in the lungs and liver of mice. Exposure of piperine before the administration of toxicant, for 7 days, substantially counteracted the levels of 8-oxo-2'-deoxyguanosine and the percentage of genomic damage in the comet assay. In a separate report, they also showed the mitigating effect of piperine on the antioxidant defense system by assessing biochemical parameters such as lipid peroxides, superoxide dismutase, catalase, and reduced glutathione (Sehgal et al. 2012). In 2013, it was further demonstrated that piperine causes a reduction in the activities of ethoxyresorufin-*o*-deethylase (specific marker for liver CYP1A1), pentoxyresorufin *o*-depentylase (specific marker for liver CYP1B1), and levels of benzo(alpha)pyrene-diol epoxide DNA adducts (Sehgal et al. 2013). These studies suggested that piperine could be a good genoprotectant against carcinogenic compounds via modulating xenobiotic-metabolizing enzymes and free radical scavenging properties.

Recently, in our laboratory, we examined the activity of piperine against cyclophosphamide in a rodent model. We observed that cyclophosphamide, 5 mg/kg, resulted in significant genomic damage and tissue-specific toxicity. Genomic damage was observed in high-proliferating tissues like bone marrow cells as well as in vital tissues like the liver and hippocampus. Interestingly, piperine treatment at the dose of 50 mg/kg for 28 days could efficiently reverse genomic damage and tissue-specific toxicities induced by cyclophosphamide. Genomic damage was quantified as the frequency of micronucleus polychromatic erythrocytes in bone marrow erythrocytes, while in hepatocytes and hippocampus, the percentage of DNA fragmentation was measured. The mechanistic approach was deciphered by evaluating the biomarkers of genotoxicity. 8-OHdG and γ H2AX are the hallmarks of genotoxicity, where 8-hydroxy-2'-deoxyguanosine (8-OHdG) is a biomarker used to detect oxidative DNA base damage provoked by reactive oxygen species while γ H2AX is a sensitive and early indicator of DNA double-strand breaks, which plays a critical role in DNA damage response. Piperine treatment reduced the levels of 8-OHdG as well as suppressed the expression of γ H2AX in hepatocytes and hippocampus, as compared to cyclophosphamide-treated rats. Further, piperine treatment also reversed the oxidative status of tissues toward normal, as presented by a reduction in reactive oxygen species formation and malondialdehyde levels, while enhancing the levels of glutathione, thus restoring the redox homeostasis disturbed as a result of cyclophosphamide treatment. On histopathological lesions in the liver and hippocampus, we observed that cyclophosphamide treatment resulted in dreadful lesions, such as a large number of apoptotic cells in both tissues, while several other specific histological lesions in the liver such as steatosis, blood filled in sinusoidal space, inflammation of cells, and dilatation of central vein with hyperemia. In subregions of the hippocampus (CA1 and DG), cyclophosphamide induced high pyknotic cells along with loss of neuronal layer in both the subregions. Such lesions are also observed in clinical scenarios with patients receiving cyclophosphamide in their chemotherapeutic regimen suggesting that our model could mimic the clinical condition. The effects were efficiently counteracted by piperine, with better reversal observed in liver cells (Yadav et al. 2023a, b). We also investigated *Piper longum* extract under similar conditions, where the extract has provided protection against cyclophosphamide-associated genomic injury and the phytochemical investigation of the extract demonstrated the presence of piperine, as one of its bioactive compounds (Yadav et al. 2023a, b).

The studies conducted in our laboratory clearly indicated that piperine treatment can reverse the DNA damage in high proliferating tissues like bone marrow, reduce tissue-specific genomic injury, diminish the markers of oxidative damage and DNA double-strand breaks, and restore the redox balance while providing protection against cytoarchitectural damage induced by cyclophosphamide, a routinely used chemotherapeutic. These findings are providing evidence to translate and harness the utilization of phytochemicals like piperine from the preclinical to the clinical testing phase.

The preventive effect of piperine is also beneficial as phytobiotics to chicken feed. da Silva Cardoso et al. (2016) performed an experiment where piperine at the

dose of 60 mg/kg was fed to 9 days old male chicks, in combination with aflatoxin B1 at the dose of 0.5 mg/kg. They analyzed the peripheral blood at various time intervals, such as 24, 48, 72, and 96 h on the first day of the experiment as well as on the last day of the experiment (on the 26th day), which was evaluated by comet assay and micronuclei formation. Animals exposed to aflatoxins for an extended period of time developed lesions resembling aflatoxicosis, featuring genomic injuries and problems with bone marrow blood production. The findings suggested that the addition of piperine to chicks after aflatoxin intoxication diminished the genomic damage and cytotoxic effects. However, future studies must replicate the effect on a larger sample size.

15.6.5 Lobeline

It is a lipophilic, pyridine alkaloid obtained from *Lobelia inflata*, which is also referred to as *Rapuntiuminflatum* or Indian weed. The pharmaceutical formulation of lobeline is commercialized as tablet preparation for use as a smoking cessation aid, despite the lack of scientific data to back up this claim. Pharmacological evidence suggests the direct interaction between lobeline and nicotinic acetylcholine receptor. It suppresses psychostimulant-associated action via modulation at the nicotinic acetylcholine receptor, along with the capability to modify presynaptic dopamine storage and release, via the interaction with vesicular monoamine transporter-2 (VMAT2) (Dwoskin and Crooks 2002). The team of Jaqueline Nascimento Picada has studied the effect of lobeline in ethanol-associated genomic instability and pilocarpine-induced seizure model. Lobeline administration to mice, at the doses of 5/10/20 mg/kg, did not change the ethanol-associated genotoxicity in peripheral blood, while preventing pilocarpine-associated DNA damage in the hippocampus, as revealed in the comet assay. The result was hypothesized to be caused by a decrease in oxidative injury, alternatively, the inhibitory action of lobeline on glutamatergic excitotoxicity may have suppressed the production of free radicals, hence lowering genomic damage (da Costa e Silva et al. 2014, 2018).

15.6.6 Vinblastine

Vincristine and vinblastine are the natural vinca alkaloid, extracted from *Vinca rosea*, and are frequently utilized as single or in conjugation with other chemotherapeutics or radiations for several malignancies. These classes of alkaloids are aneugens (producing anomalous chromosome number) and mitotic spindle poison, they associate with tubulin and suppress the gathering of microtubules. Aneugens are predicted to possess a lower degree of clastogenicity because they prevent the development of the mitotic spindle. However, vincristine is reported to augment genomic damage; vinblastine, on the other hand, demonstrated defense to normal cell gamma-radiation-associated genome injury. Rajagopalan et al. (2003) demonstrated that vinblastine when administered to mice, before gamma-radiation

contact, suppressed genome injury in fibrosarcoma, leukocytes, and bone marrow cells, as revealed through comet assay. The action was postulated due to the differential feature of cancer and non-cancer cells, where vinblastine condensed mitosis in cancer cells, which has its accessibility to interact with the genome, thereby preventing radiation-induced damage.

15.6.7 Vincamine

Vincamine is a monoterpenoid indole alkaloid extracted from the leaves of *Vinca minor*. Clinically, it is widely utilized in Europe, as a prescription medicine for the treatment of primary degenerative and vascular dementia. Vincamine is also demonstrated to possess multiple biological activities like antioxidant, anti-inflammatory, and antiapoptotic characteristics. In a very recent study, prior oral administration of vincamine, at the dose of 40 mg/kg for 7 consecutive days, with a single exposure of cisplatin to rats, prevented genomic injury in renal tissue. Vincamine antioxidant and anti-inflammatory properties are suggested to be the cause of this effect, which was achieved by engaging the Nrf2/HO-1 pathway and preventing the TLR4-IFN- γ -CD44 cell's inflammatory state (El-Sayed et al. 2021).

15.6.8 Trigonelline

Chemically, trigonelline is *N*-methyl nicotinic acid ($C_7H_7NO_2$), a heterocyclic alkaloid, which is extracted from *Trigonella foenum-graecum* (fenugreek); from the family, Fabaceae. It offers a wide range of therapeutic benefits and is widely known to treat conditions like diabetes mellitus and neurological disorders (Zhou et al. 2012). A detailed analysis of trigonelline was conducted by Tanveer et al. (2023), where its genome protective action was established against ultraviolet B radiation using human dermal fibroblasts and mice models. The effect was demonstrated to be regulated by main hallmarks of DNA damage consisting of ATM (ataxia-telangiectasia mutated), ATR (ATM- and Rad3-Related), p53, Gamma-H2AX (H2A histone family member X), and the Checkpoint kinases (Chk1 and Chk2). Further, trigonelline treatment also modulated the PI3K-AKT-Nrf2 signaling pathway and offered genomic stability. The study suggests that this property can be utilized in the cosmeceutical industry to guard dermis cells against exposure to ultraviolet damage.

15.6.9 Thebaine

Thebaine, also known as paramorphine, is an isoquinoline derivative and an alkaloid that is isolated from the dried latex of immature *Papaver somniferum*, often known as the opium poppy or bread seed poppy and is a member of the Papaveraceae family. In 2018, the European Food Safety Authority arrived at the decision that

there was insufficient evidence to carry out a quantitative risk categorization of thebaine (Eisenreich et al. 2020). Reimann et al. (2021) showed that a high dose of thebaine induced micronucleus formation and no effects were observed in the comet assay; however, when co-exposed with mitomycin-C, a reduction in micronucleus frequency was observed in TK6 cells. Thus, further analysis is required to conclude the findings.

15.6.10 Gramine

Gramine is an indole nucleus containing alkaloid, also referred to as donaxine, and can be isolated from a variety of plants like *Avenasativa* (commonly known as oat, from family Gramineae), *Hordeum vulgare*, *Triticumaestivum*, etc. The pharmacological properties of gramine are comparable to those of ephedrine and include the relaxation of bronchial smooth muscle, vasorelaxation, and the treatment of bronchitis, nephritis, and bronchial asthma. Pathak et al. (2018) conducted various assays to evaluate the genotoxicity of gramine under in vivo conditions. Bacterial mutagenicity was tested in *S. typhimurium* strain, where gramine (at the concentration range of 1250–10,000 µg/plate) did not induce any mutagenic response, while in the presence of a metabolizing enzyme, gramine co-exposed with sodium azide, attenuated the colonies in TA1535 strains. Also, when administered to mice at the dose of 150 mg/kg, the results of genotoxicity were negative as determined by micronucleus polychromatic erythrocyte and chromosomal aberrations. However, this evidence is not comprehensive and needs to be assessed in various cell culture/preclinical models for detailed mechanistic exploration.

15.6.11 Beta-Carbolines Alkaloid

Beta-carboline alkaloids are indole alkaloid derivatives, with a major effect on cerebrum functioning, chemically containing tricyclicpyrido[3,4-b] indole cycle. Moura et al. (2007) investigated the action of aromatic (harmine, harmine, and harmol) and dihydro-beta-carboline (harmaline and harmalol) alkaloids for genoprotection in Chinese hamster lung fibroblasts. They observed that harmine, harmine, and harmol were able to produce genomic damage only at the concentration of 40 µg/mL, no damage was observed at 10 and 20 µg/mL, as calculated in the comet assay, while harmaline and harmalol did not induce any damage at the same concentration. However, both aromatic and dihydro-beta-carbolines showed a reduction in DNA damage index and damage frequency when treated against 100 µM concentration of hydrogen peroxide. These reports point toward the genoprotective potential of beta-carboline, where the action is attributed to the hydroxyl radical-scavenging property; however, in vivo investigations are required to validate these findings.

15.6.12 Rutaecarpine

Rutecarpine or rutaecarpine, known for potent cyclooxygenase-2 inhibition, is derived from *Tetradiumruticarpum* and categorized as a nonbasic alkaloid. Li et al. (2019) investigated the action of rutaecarpine against benzo[a] pyrene-associated genotoxic stress. When rutaecarpine, at the concentration of 10 μM , was exposed in combination with benzo[a]pyrene (5 μM), to mucoepidermoid pulmonary carcinoma cells and hepatocellular carcinoma cells, a significant reduction in genomic injury was observed. The effect was examined by ethoxyresorufin-*O*-deethylase assay and Muse™Multi-Color DNA Damage Kit. The findings suggested that rutaecarpine plays a significant role in controlling the cytochrome activity of hepatocytes and lung cancer cells.

15.6.13 Aeroplysinin-1 and Aaptamine

Marine organisms are an incredibly rich source of small molecules with pharmacological properties that may be good for human health. Marine sponges, belonging to phylum Poriphera, especially of Aplysina genus, serve as a reservoir for brominated isoxazoline alkaloids, amid which aeroplysinin-1, isolated from *Aplysiniacavernicola*, is a very well-known anti-inflammatory, anti-tumor, and anti-angiogenic compound (García-Vilas et al. 2015). Similarly, Genus *Aaptos*, is also a reservoir for a bioactive benzo[de][1,6]-naphthyridine alkaloids, known for anticancer and antiviral properties, is referred to as aaptamine (Nadar et al. 2022). Funk et al. (2014), established the genoprotective property of aaptamine and aeroplysinin-1 in rat renal tubular cells and rat glomerular endothelial cells against cisplatin. They confirmed the alteration in DNA adduct formation and nuclear gamma H2AX expression in pretreated cells with aaptamine and aeroplysinin-1, at the concentration of 2 $\mu\text{mol/L}$, prior to cisplatin exposure (30 $\mu\text{mol/L}$). Similar observations were presented by Stuhldreier et al. (2015), against daunorubicin (0.12 μM) in acute myeloid leukemia, where aaptamine and aeroplysinin-1 diminished gamma H2AX expression, phosphorylation of KRAB domain protein (KAP1), and phosphorylation of checkpoint kinase 1 (Chk1). Thus, both studies provide conclusive results that aaptamine and aeroplysinin-1 protect against genotoxic stress and should be explored in further experiments. Also, such activities might be explored to broaden the therapeutic window of chemotherapeutic agents like cisplatin and daunorubicin, consequently rescuing the cells from associated side effects.

15.7 Potential Problems in Using Alkaloids

The above literature is suggesting that alkaloids have the potential to mitigate the genotoxic sequelae specifically caused by chemotherapeutic drugs. Despite the large number of studies that have been done and published about the advantages of alkaloids or phytochemicals in chemotherapy, there remain numerous critics and

skeptics among medical professionals and patients alike. This may be the result of ignorance on the one hand, but the spread of myths and erroneous beliefs is also a significant factor. However, the toxicity of many alkaloids remains another critical issue when translating their use to clinical scenarios. The primary function of alkaloids in flora is generally acknowledged to be toxicity toward predators and diseases. Alkaloids like caffeine, which is consumed to increase mental clarity; nicotine, to improve physical performance; morphine, a common analgesic; codeine, which is used to treat coughs; and strychnine, a potent tetanic toxin, are known to have serious side effects or toxic effects. Alkaloids are further known to demonstrate organ-specific toxicity (such as neuro-, cardio-, hepato-, and nephrotoxicity), while the mechanistic pathways for such toxicity appear to be associated with enzymatic modifications affecting physiological progressions, respiratory suppression, interaction/alkylating with genome and proteins, prevention of DNA synthesis and repair pathways, suppression of cell division, etc. However, they are also known to possess pleiotropic nature where they show toxic as well as preventive potential as demonstrated in cell line and in vivo models (Matsuura and Fett-Neto 2015). In a similar context, their capacity to interact with the genome has been thoroughly investigated. The majority of these chemicals exhibit DNA-damaging or cytotoxic effects and are consistent with genome interaction. They are shown to interact with DNA via highly specific non-covalent, reversible, and formation of molecular complexes with the genome (Basu and Kumar 2018; Sjakste et al. 2020). Likewise, chemotherapeutic medicines are designed to target DNA and this could result in possible interactions with alkaloids; hence, careful examination and study design is required to draw any conclusions.

15.8 Conclusion

The present chapter has provided a glimpse of major sources and signaling pathways involved in mediating genotoxic stress. However, genotoxic stress can also be exaggerated by several other pathophysiological processes and secondary consequences of genotoxins (such as oxidative stress and inflammatory response) (George et al. 2017; Cristóbal-Luna et al. 2018; Yubo et al. 2021). Therefore, an understanding of the molecular mechanisms by which genotoxic stress is progressed in physiology is required to produce effective therapies with fewer adverse effects. In the case of alkaloids, studies clearly indicate that alkaloids are efficient in reversing genotoxic stress as demonstrated in cell culture and rodent models. Experimental data from in vitro and in vivo studies suggest that alkaloids have been able to combat various biomarkers and cellular responses against DNA damage. However, there are also a few studies that contradict the same. To avoid undesired consequences such as increasing genomic injury via oxidative stress, alterations in autophagy to remove toxic biomolecules, and/or inhibition of critical pro-survival trails to cell stress, it is of great significance to cautiously characterize the hormetic impact of these alkaloids. Further, most of the experiments are often limited in providing mechanistic approaches to these compounds, where the studies did not reveal the target

signaling pathway of DNA damage or provide very little information on DNA repair machinery. Additionally, there is negligible to limited evidence from clinical trials with controlled human interventions, where alkaloids are showing effectiveness against genotoxic stress. While the comparatively short period of these interventions may have contributed to the lack of impact of alkaloids on DNA stability in human investigations, it is also possible that these chemicals, even at dietary-appropriate amounts, are ineffective genoprotectants in vivo. Thus, although findings across the globe are showing encouragement, more studies must be conducted emphasizing mechanistic approaches and clinical translation of these compounds.

References

- Aprotosoaie AC, Luca VS, Trifan A, Miron A (2019) Antigenotoxic potential of some dietary non-phenolic phytochemicals. *Stud Nat Prod Chem* 60:223–297. <https://doi.org/10.1016/B978-0-444-64181-6.00007-3>
- Balakrishnan S, Vellaichamy L, Menon VP, Manoharan S (2008) Antigenotoxic effects of curcumin and piperine alone or in combination against 7,12-dimethylbenz(a)anthracene induced genotoxicity in bone marrow of Golden Syrian hamsters. *Toxicol Mech Methods* 18(9): 691–696. <https://doi.org/10.1080/15376510701781520>, PMID 20020926
- Bartek J, Lukas J (2007) DNA damage checkpoints: from initiation to recovery or adaptation. *Curr Opin Cell Biol* 19(2):238–245. <https://doi.org/10.1016/j.ceb.2007.02.009>, PMID 17303408
- Basu A, Kumar GS (2018) Nucleic acids binding strategies of small molecules: lessons from alkaloids. *Biochim Biophys Acta Gen Subj* 1862(9):1995–2016. <https://doi.org/10.1016/j.bbagen.2018.06.010>, PMID 29908208
- Caetano BFR, Tablas MB, Pereira NEF, de Moura NA, Carvalho RF, Rodrigues MAM et al (2018) Capsaicin reduces genotoxicity, colonic cell proliferation and preneoplastic lesions induced by 1,2-dimethylhydrazine in rats. *Toxicol Appl Pharmacol* 338:93–102. <https://doi.org/10.1016/j.taap.2017.11.008>, PMID 29155087
- Castillo WO, Aristizabal-Pachon AF (2017) Galantamine protects against beta amyloid peptide-induced DNA damage in a model for Alzheimer's disease. *Neural Regen Res* 12(6):916–917. <https://doi.org/10.4103/1673-5374.208572>, PMID 28761423
- Castillo WO, Aristizabal-Pachon AF, de Lima Montaldi AP, Sakamoto-Hojo ET, Takahashi CS (2016) Galanthamine decreases genotoxicity and cell death induced by β -amyloid peptide in SH-SY5Y cell line. *Neurotoxicology* 57:291–297. <https://doi.org/10.1016/j.neuro.2016.10.013>, PMID 27793617
- Castillo WO, Aristizabal-Pachon AF, Sakamoto-Hojo E, Gasca CA, Cabezas-Fajardo FA, Takahashi C (2018) Caliphurria subdentata (Amaryllidaceae) decreases genotoxicity and cell death induced by β -amyloid peptide in SH-SY5Y cell line. *Mutat Res Genet Toxicol Environ Mutagen* 836(Pt B):54–61. <https://doi.org/10.1016/j.mrgentox.2018.06.010>, PMID 30442346
- Chatterjee N, Walker GC (2017) Mechanisms of DNA damage, repair, and mutagenesis. *Environ Mol Mutagen* 58(5):235–263. <https://doi.org/10.1002/em.22087>, PMID 28485537
- Coates PJ, Lorimore SA, Wright EG (2005) Cell and tissue responses to genotoxic stress. *J Pathol* 205(2):221–235. <https://doi.org/10.1002/path.1701>, PMID 15643669
- Cooke MS, Evans MD, Dizdaroğlu M, Lunec J (2003) Oxidative DNA damage: mechanisms, mutation, and disease. *FASEB J* 17(10):1195–1214. <https://doi.org/10.1096/fj.02-0752rev>, PMID 12832285
- Cristóbal-Luna JM, Álvarez-González I, Madrigal-Bujaidar E, Chamorro-Cevallos G (2018) Grapefruit and its biomedical, antigenotoxic and chemopreventive properties. *Food Chem Toxicol* 112:224–234. <https://doi.org/10.1016/j.fct.2017.12.038>, PMID 29284137

- da Costa E Silva LD, Rodrigues LC, Dos Santos VR, da Costa AM, de Barros Falcão Ferraz A, Rolla HC et al (2014) Evaluation of mutagenic and genotoxic activities of lobeline and its modulation on genomic instability induced by ethanol. *Life Sci* 103(2):73–78. <https://doi.org/10.1016/j.lfs.2014.03.034>, PMID 24727238
- da Costa E Silva LD, Pereira P, Regner GG, Boaretto FBM, Hoffmann C, Pflüger P et al (2018) DNA damage and oxidative stress induced by seizures are decreased by anticonvulsant and neuroprotective effects of lobeline, a candidate to treat alcoholism. *Metab Brain Dis* 33(1): 53–61. <https://doi.org/10.1007/s11011-017-0130-1>, PMID 29032429
- da Silva Cardoso V, Vermelho AB, Ribeiro de Lima CA, Mendes de Oliveira J, Freire de Lima ME, Pinto da Silva LH et al (2016) Antigenotoxic effect of piperine in broiler chickens intoxicated with aflatoxin b1. *Toxins* 8(11):316. <https://doi.org/10.3390/toxins8110316>, PMID 27809242
- de Kok TM, de Waard P, Wilms LC, van Breda SG (2010) Antioxidative and antigenotoxic properties of vegetables and dietary phytochemicals: the value of genomics biomarkers in molecular epidemiology. *Mol Nutr Food Res* 54(2):208–217. <https://doi.org/10.1002/mnfr.200900288>, PMID 20024933
- Debnath B, Singh WS, Das M, Goswami S, Singh MK, Maiti D et al (2018) Role of plant alkaloids on human health: a review of biological activities. *Mater Today Chem* 9:56–72. <https://doi.org/10.1016/j.mtchem.2018.05.001>
- Dey P, Kundu A, Kumar A, Gupta M, Lee BM, Bhakta T et al (2020) Analysis of alkaloids (indole alkaloids, isoquinoline alkaloids, tropane alkaloids). In: *Recent advances in natural products analysis*. Elsevier, Amsterdam, pp 505–567
- Díaz Barriga Arceo S, Madrigal-Bujaidar E, Calderón Montellano E, Ramírez Herrera L, Díaz García BD (1995) Genotoxic effects produced by capsaicin in mouse during subchronic treatment. *Mutat Res Genet Toxicol* 345(3–4):105–109. [https://doi.org/10.1016/0165-1218\(95\)90046-2](https://doi.org/10.1016/0165-1218(95)90046-2)
- Dwoskin LP, Crooks PA (2002) A novel mechanism of action and potential use for lobeline as a treatment for psychostimulant abuse. *Biochem Pharmacol* 63(2):89–98. [https://doi.org/10.1016/s0006-2952\(01\)00899-1](https://doi.org/10.1016/s0006-2952(01)00899-1), PMID 11841781
- Eisenreich A, Sachse B, Gürtler R, Dusemund B, Lindtner O, Schäfer B (2020) What do we know about health risks related to thebaine in food? *Food Chem* 309:125564. <https://doi.org/10.1016/j.foodchem.2019.125564>, PMID 31670127
- El-Sayed RM, Abo El Gheit RE, Badawi GA (2021) Vincamine protects against cisplatin induced nephrotoxicity via activation of Nrf2/HO-1 and hindering TLR4/IFN- γ /CD44 cells inflammatory cascade. *Life Sci* 272:119224. <https://doi.org/10.1016/j.lfs.2021.119224>, PMID 33610575
- Figueroa-González G, Pérez-Plasencia C (2017) Strategies for the evaluation of DNA damage and repair mechanisms in cancer. *Oncol Lett* 13(6):3982–3988. <https://doi.org/10.3892/ol.2017.6002>, PMID 28588692
- Firsanov DV, Kulikova VA, Solovjeva LV, Mikhailov VM, Nikiforov AA, Svetlova MP (2021) Methods for the detection of DNA damage. In: *Genome stability*. Academic, New York, pp 679–697
- Funk F, Krüger K, Henninger C, Wätjen W, Proksch P, Thomale J et al (2014) Spongean alkaloids protect rat kidney cells against cisplatin-induced cytotoxicity. *Anti-Cancer Drugs* 25(8): 917–929. <https://doi.org/10.1097/CAD.000000000000119>, PMID 24820908
- García-Vilas JA, Martínez-Poveda B, Quesada AR, Medina MÁ (2015) Aeropylsinin-1, a sponge-derived multi-targeted bioactive marine drug. *Mar Drugs* 14(1):1. <https://doi.org/10.3390/md14010001>, PMID 26703630
- George VC, Dellaire G, Rupasinghe HPV (2017) Plant flavonoids in cancer chemoprevention: role in genome stability. *J Nutr Biochem* 45:1–14. <https://doi.org/10.1016/j.jnutbio.2016.11.007>, PMID 27951449
- Georgoulis A, Vorgias CE, Chrousos GP, Rogakou EP (2017) Genome instability and γ H2AX. *Int J Mol Sci* 18(9):1979. <https://doi.org/10.3390/ijms18091979>, PMID 28914798

- Geriyol P, Basavanneppa HB, Dhananjaya BL (2015) Protecting effect of caffeine against vinblastine (an anticancer drug) induced genotoxicity in mice. *Drug Chem Toxicol* 38(2):188–195. <https://doi.org/10.3109/01480545.2014.928720>, PMID 24956935
- Hosseinimehr SJ (2010) Flavonoids and genomic instability induced by ionizing radiation. *Drug Discov Today* 15(21–22):907–918. <https://doi.org/10.1016/j.drudis.2010.09.005>, PMID 20933097
- Izquierdo-Vega JA, Morales-González JA, Sánchez-Gutiérrez M, Betanzos-Cabrera G, Sosa-Delgado SM, Sumaya-Martínez MT et al (2017) Evidence of some natural products with antigenotoxic effects. Part 1: fruits and polysaccharides. *Nutrients* 9(2):102. <https://doi.org/10.3390/nu9020102>, PMID 28157162
- Kanehisa M, Goto S, Kawashima S, Okuno Y, Hattori M (2004) The KEGG resource for deciphering the genome. *Nucleic Acids Res* 32(Database issue):D277–D280. <https://doi.org/10.1093/nar/gkh063>, PMID 14681412
- Kondo N, Takahashi A, Ono K, Ohnishi T (2010) DNA damage induced by alkylating agents and repair pathways. *J Nucleic Acids* 2010:543531. <https://doi.org/10.4061/2010/543531>, PMID 21113301
- Kovalchuk I (2021) Cancer and genomic instability. In: *Genome stability*. Academic, New York, pp 495–519
- Kumar SS, Devasagayam TP, Jayashree B, Kesavan PC (2001) Mechanism of protection against radiation-induced DNA damage in plasmid pBR322 by caffeine. *Int J Radiat Biol* 77(5):617–623. <https://doi.org/10.1080/09553000110034649>, PMID 11382340
- Lee SJ, Ahn H, Nam KW, Kim KH, Mar W (2012) Effects of rutaecarpine on hydrogen peroxide-induced apoptosis in murine hepa-1c1c7 cells. *Biomol Ther (Seoul)* 20(5):487–491. <https://doi.org/10.4062/biomolther.2012.20.5.487>, PMID 24009839
- Lehmann F, Wennerberg J (2021) Evolution of nitrogen-based alkylating anticancer agents. *Processes* 9(2):377. <https://doi.org/10.3390/pr9020377>
- Lewinska A, Jarosz P, Czech J, Rzeszutek I, Bielak-Zmijewska A, Grabowska W et al (2015) Capsaicin-induced genotoxic stress does not promote apoptosis in A549 human lung and DU145 prostate cancer cells. *Mutat Res Genet Toxicol Environ Mutagen* 779:23–34. <https://doi.org/10.1016/j.mrgentox.2015.02.003>, PMID 25813723
- Li L, Zou L (2005) Sensing, signaling, and responding to DNA damage: organization of the checkpoint pathways in mammalian cells. *J Cell Biochem* 94(2):298–306. <https://doi.org/10.1002/jcb.20355>, PMID 15578575
- Li Y, Park MS, Chan WK, Livesey J, Zhou Z (2019) Effect of rutaecarpine on the benzo(a)pyrene-induced DNA damage in vitro. *FASEB J* 33(Suppl 1):672–679. https://doi.org/10.1096/fasebj.2019.33.1_supplement.672.9
- Liu YQ, Wang XL, He DH, Cheng YX (2021) Protection against chemotherapy- and radiotherapy-induced side effects: a review based on the mechanisms and therapeutic opportunities of phytochemicals. *Phytomedicine* 80:153402. <https://doi.org/10.1016/j.phymed.2020.153402>, PMID 33203590
- López-Romero D, Izquierdo-Vega JA, Morales-González JA, Madrigal-Bujaidar E, Chamorro-Cevallos G, Sánchez-Gutiérrez M et al (2018) Evidence of some natural products with antigenotoxic effects. Part 2: plants, vegetables, and natural resin. *Nutrients* 10(12):1954. <https://doi.org/10.3390/nu10121954>, PMID 30544726
- Luca VS, Miron A, Aprotosoaie AC (2016) The antigenotoxic potential of dietary flavonoids. *Phytochem Rev* 15(4):591–625. <https://doi.org/10.1007/s11101-016-9457-1>
- Matsuura HN, Fett-Neto AG (2015) Plant alkaloids: main features, toxicity, and mechanisms of action. *Plant Toxins* 2(7):1–5
- Mohamed SA, Upreti S, Rajendra SV, Dang R (2017) Genotoxicity: mechanisms, testing guidelines and methods. *Glob J Pharm Pharm Sci* 1(5):133–138
- Mohan G, Das A, Rajyaguru PI (2021) Genotoxic stress response: what is the role of cytoplasmic mRNA fate? *BioEssays* 43(8):e2000311. <https://doi.org/10.1002/bies.202000311>, PMID 34096096

- Morales-González JA, Madrigal-Bujaidar E, Sánchez-Gutiérrez M, Izquierdo-Vega JA, Valadez-Vega MDC, Álvarez-González I et al (2019) Garlic (*Allium sativum* L.): a brief review of its antigenotoxic effects. *Foods* 8(8):343. <https://doi.org/10.3390/foods8080343>, PMID 31412555
- Moura DJ, Richter MF, Boeira JM, Pêgas Henriques JA, Saffi J (2007) Antioxidant properties of β -carboline alkaloids are related to their antimutagenic and antigenotoxic activities. *Mutagenesis* 22(4):293–302. <https://doi.org/10.1093/mutage/gem016>, PMID 17545209
- Nadar VM, Manivannan S, Chinnaiyan R, Govarthanan M, Ponnuchamy K (2022) Review on marine sponge alkaloid, aaptamine: a potential antibacterial and anticancer drug. *Chem Biol Drug Des* 99(1):103–110. <https://doi.org/10.1111/cbdd.13932>, PMID 34331335
- Nehlig A, Debry G (1994) Potential genotoxic, mutagenic and antimutagenic effects of coffee: a review. *Mutat Res* 317(2):145–162. [https://doi.org/10.1016/0165-1110\(94\)90022-1](https://doi.org/10.1016/0165-1110(94)90022-1), PMID 7511793
- Nelson BC, Wright CW, Ibuki Y, Moreno-Villanueva M, Karlsson HL, Hendriks G et al (2017) Emerging metrology for high-throughput nanomaterial genotoxicology. *Mutagenesis* 32(1): 215–232. <https://doi.org/10.1093/mutage/gew037>, PMID 27565834
- Pathak MP, Policegoudra RS, Goyary D, Das A, Mandal S, Chakraborti S et al (2018) Safety evaluation of an oat grain alkaloid gramine by genotoxicity assays. *Drug Chem Toxicol* 41(2): 147–154. <https://doi.org/10.1080/01480545.2017.1322605>, PMID 28523965
- Poirier MC (2012) Chemical-induced DNA damage and human cancer risk. *Discov Med* 14(77): 283–288. <https://doi.org/10.1038/nrc1410>, PMID 23114584
- Proshkina E, Shaposhnikov M, Moskalev A (2020) Genome-protecting compounds as potential geroprotectors. *Int J Mol Sci* 21(12):4484. <https://doi.org/10.3390/ijms21124484>, PMID 32599754
- Rajagopalan R, Ranjan SK, Nair CK (2003) Effect of vinblastine sulfate on γ -radiation-induced DNA single-strand breaks in murine tissues. *Mutat Res* 536(1–2):15–25. [https://doi.org/10.1016/S1383-5718\(03\)00015-9](https://doi.org/10.1016/S1383-5718(03)00015-9), PMID 12694742
- Raymond P, Thompson C, Longstaff E (2004) Examination of the potential genotoxicity of pure capsaicin in bacterial mutation chromosome aberration and rodent micronucleus tests. *Environ Mol Mutagen* 44(5):441–447. 10.1002/(ISSN)1098-2280. 10.1002/em.v44:5 <https://doi.org/10.1002/em.20072>
- Reddig A, Rübe CE, Rödiger S, Schierack P, Reinhold D, Roggenbuck D (2018) DNA damage assessment and potential applications in laboratory diagnostics and precision medicine. *J Lab Precis Med* 3:31. <https://doi.org/10.21037/jlpm.2018.03.06>
- Reen RK, Wiebel FJ, Singh J (1997) Piperine inhibits aflatoxin b1-induced cytotoxicity and genotoxicity in V79 Chinese hamster cells genetically engineered to express rat cytochrome P4502B1. *J Ethnopharmacol* 58(3):165–173. [https://doi.org/10.1016/S0378-8741\(97\)00104-9](https://doi.org/10.1016/S0378-8741(97)00104-9), PMID 9421252
- Reimann H, Bankoglu EE, Stopper H, Hintzsche H (2021) In vitro evaluation of chromosomal damage and DNA strand breaks after treatment with the poppy seed alkaloid thebaine. *Mutat Res Genet Toxicol Environ Mutagen* 870–871:503393. <https://doi.org/10.1016/j.mrgentox.2021.503393>, PMID 34583825
- Richeux F, Cascante M, Ennamany R, Sanchez D, Sanni A, Saboureau D et al (2000) Implications of oxidative stress and inflammatory process in the cytotoxicity of capsaicin in human endothelial cells: lack of DNA strand breakage. *Toxicology* 147(1):41–49. [https://doi.org/10.1016/S0300-483X\(00\)00184-0](https://doi.org/10.1016/S0300-483X(00)00184-0), PMID 10837931
- Sak K (2012) Chemotherapy and dietary phytochemical agents. *Chemother Res Pract* 2012:282570. <https://doi.org/10.1155/2012/282570>, PMID 23320169
- Sehgal A, Kumar M, Jain M, Dhawan DK (2011) Combined effects of curcumin and piperine in ameliorating benzo(a)pyrene induced DNA damage. *Food Chem Toxicol* 49(11):3002–3006. <https://doi.org/10.1016/j.fct.2011.07.058>, PMID 21827816
- Sehgal A, Kumar M, Jain M, Dhawan DK (2012) Synergistic effects of piperine and curcumin in modulating benzo(a)pyrene induced redox imbalance in mice lungs. *Toxicol Mech Methods* 22(1):74–80. <https://doi.org/10.3109/15376516.2011.603392>, PMID 21859361

- Sehgal A, Kumar M, Jain M, Dhawan DK (2013) Modulatory effects of curcumin in conjunction with piperine on benzo(a)pyrene-mediated DNA adducts and biotransformation enzymes. *Nutr Cancer* 65(6):885–890. <https://doi.org/10.1080/01635581.2013.805421>, PMID 23909733
- Selvendiran K, Padmavathi R, Magesh V, Sakthisekaran D (2005) Preliminary study on inhibition of genotoxicity by piperine in mice. *Fitoterapia* 76(3–4):296–300. <https://doi.org/10.1016/j.fitote.2005.03.016>, PMID 15890459
- Shiloh Y (2006) The ATM-mediated DNA-damage response: taking shape. *Trends Biochem Sci* 31(7):402–410. <https://doi.org/10.1016/j.tibs.2006.05.004>, PMID 16774833
- Sirbu BM, Cortez D (2013) DNA damage response: three levels of DNA repair regulation. *Cold Spring Harb Perspect Biol* 5(8):a012724. <https://doi.org/10.1101/cshperspect.a012724>, PMID 23813586
- Sjakste N, Djelić N, Dzintare M, Živković L (2020) LDNA-BINDING and DNA-protecting activities of small natural organic molecules and food extracts. *Chem Biol Interact* 323: 109030. <https://doi.org/10.1016/j.cbi.2020.109030>, PMID 32205154
- Stoilov LM, Mullenders LH, Natarajan AT (1994) Caffeine potentiates or protects against radiation-induced DNA and chromosomal damage in human lymphocytes depending on temperature and concentration. *Mutat Res* 311(2):169–174. [https://doi.org/10.1016/0027-5107\(94\)90174-0](https://doi.org/10.1016/0027-5107(94)90174-0), PMID 7526181
- Stuhldreier F, Kassel S, Schumacher L, Wesselborg S, Proksch P, Fritz G (2015) Pleiotropic effects of spongean alkaloids on mechanisms of cell death, cell cycle progression and DNA damage response (DDR) of acute myeloid leukemia (AML) cells. *Cancer Lett* 361(1):39–48. <https://doi.org/10.1016/j.canlet.2015.02.030>, PMID 25697484
- Swift LH, Golsteyn RM (2014) Genotoxic anti-cancer agents and their relationship to DNA damage, mitosis, and checkpoint adaptation in proliferating cancer cells. *Int J Mol Sci* 15(3): 3403–3431. <https://doi.org/10.3390/ijms15033403>, PMID 24573252
- Tanveer MA, Rashid H, Nazir LA, Archoo S, Shahid NH, Ragni G et al (2023) Trigonelline, a plant derived alkaloid prevents ultraviolet-B-induced oxidative DNA damage in primary human dermal fibroblasts and BALB/c mice via modulation of phosphoinositide 3-kinase-Akt-Nrf2 signalling axis. *Exp Gerontol* 171:112028. <https://doi.org/10.1016/j.exger.2022.112028>, PMID 36384201
- Turkez H, Arslan ME, Ozdemir O (2017) Genotoxicity testing: progress and prospects for the next decade. *Expert Opin Drug Metab Toxicol* 13(10):1089–1098. <https://doi.org/10.1080/17425255.2017.1375097>, PMID 28889778
- Villarroya M, García AG, Marco-Contelles J, López MG (2007) An update on the pharmacology of galantamine. *Expert Opin Investig Drugs* 16(12):1987–1998. <https://doi.org/10.1517/13543784.16.12.1987>, PMID 18042006
- Yadav V, Krishnan A, Zahiruddin S, Ahmad S, Vohora D (2023a) Amelioration of cyclophosphamide-induced DNA damage, oxidative stress, and hepato- and neurotoxicity by Piper longum extract in rats: the role of γ H2AX and 8-OHdG. *Front Pharmacol* 14:1147823. <https://doi.org/10.3389/fphar.2023.1147823>, PMID 36969834
- Yadav V, Krishnan A, Vohora D (2023b) Altered hallmarks of DNA double-strand breaks, oxidative DNA damage and cytogenotoxicity by piperlongumine in hippocampus and hepatocytes of rats intoxicated with cyclophosphamide. *Life Sci* 316:121391. <https://doi.org/10.1016/j.lfs.2023.121391>, PMID 36657641
- Yang J, Xu ZP, Huang Y, Hamrick HE, Duerksen-Hughes PJ, Yu YN (2004) ATM and ATR: sensing DNA damage. *World J Gastroenterol* 10(2):155–160. <https://doi.org/10.3748/wjg.v10.i2.155>, PMID 14716813
- Yubo L, Chen Q, Zhang N, Zhang K, Dou T, Cao Y, Liu Y, Li K, Hao X, Xie X, Li W, Ren Y, Zhang J (2021) Author correction: proteomic profiling and genome-wide mapping of O-GlcNAc chromatin-associated proteins reveal an O-GlcNAc-regulated genotoxic stress response. *Nat Commun* 12(1). <https://doi.org/10.1038/s41467-021-26886-5>

- Zhou BB, Elledge SJ (2000) The DNA damage response: putting checkpoints in perspective. *Nature* 408(6811):433–439. <https://doi.org/10.1038/35044005>, PMID 11100718
- Zhou J, Chan L, Zhou S (2012) Trigonelline: a plant alkaloid with therapeutic potential for diabetes and central nervous system disease. *Curr Med Chem* 19(21):3523–3531. <https://doi.org/10.2174/092986712801323171>, PMID 22680628